

## Digits in Disguise

Puzzles where letters or blanks hide digits. Solve them using place value, divisibility tests, and careful reasoning about each position.

### Two kinds of digit puzzles

#### 1. Cryptarithms (letters stand for digits)

- Each **letter is a single digit**, and the **same letter** always means the **same digit**.
- Usually a **leading digit is not 0**.
- Strategy: look at the **units column** first, watch the **carrying**, and use any structure (like a repeated-digit answer).

$$A B \times A = C C C \rightarrow C C C = C \times 111 = C \times 3 \times 37, \text{ so } 37 \times 3 = 111 \text{ (A=3, B=7, C=1)}$$

#### 2. Finding a missing digit for divisibility

When a number has a blank (or letter), use the relevant **divisibility test** to pin down the missing digit.

- For **9**: make the digit sum a multiple of 9.
- For **18**: the number must be divisible by **2 and 9** together.
- For **3**: make the digit sum a multiple of 3.

$$31z5 \text{ is a multiple of } 9 \Rightarrow 3 + 1 + z + 5 = 9 + z \text{ is a multiple of } 9 \Rightarrow z = 0 \text{ or } 9$$

**Prerequisite refresher:** A "repdigit" like 111, 222... equals (that digit)  $\times$  111. And  $111 = 3 \times 37$ .

### Where marks slip away

**Trap 1 — Leading zeros.** The first digit of a number can't be 0, so rule those letter values out.

**Trap 2 — List every solution.** "Find all z" often has more than one answer (e.g.  $z = 0$  or 9). Don't stop at the first.

**Trap 3 — Combine tests for composite divisors.** For 18 use 2 and 9; for 6 use 2 and 3 — both conditions must hold.

**Trap 4 — Same letter, same digit.** If C appears three times, it is the same digit each time.

**Slip 5 — Check your answer** by substituting the digit back into the original number.

## Model answers — write them like this

**Example A. Solve the cryptarithm  $A B \times A = C C C$ .**

**Step 1:** CCC is a repdigit, so  $CCC = C \times 111 = C \times 3 \times 37$ .

**Step 2:** So one factor should be 37 and the other a single digit:  $37 \times 3 = 111$ .

**Step 3:** Matching  $A B \times A$ :  $A = 3$ ,  $B = 7$ , and  $C = 1$ .

$$\therefore A = 3, B = 7, C = 1 \quad (37 \times 3 = 111)$$

**Example B. Find all digits  $z$  so that  $31z5$  is a multiple of 9.**

**Step 1:** Digit sum  $= 3 + 1 + z + 5 = 9 + z$ .

**Step 2:**  $9 + z$  must be a multiple of 9, so  $z = 0$  (gives 9) or  $z = 9$  (gives 18).

$$\therefore z = 0 \text{ or } 9$$

**Example C. Find all digits  $a$  so that  $6a2$  is divisible by 3.**

**Step 1:** Digit sum  $= 6 + a + 2 = 8 + a$ .

**Step 2:**  $8 + a$  must be a multiple of 3, so  $a = 1$  (9), 4 (12) or 7 (15).

$$\therefore a = 1, 4 \text{ or } 7$$

**Example D. Find all digits  $b$  so that  $732b$  is divisible by 6.**

**Step 1:** Divisible by 6 means divisible by 2 and by 3. For 2,  $b$  must be even.

**Step 2:** For 3, digit sum  $7 + 3 + 2 + b = 12 + b$  must be a multiple of 3, so  $b = 0, 3, 6$  or 9.

**Step 3:** The even values among those are  $b = 0$  and 6.

$$\therefore b = 0 \text{ or } 6$$

Next: [try the worksheet](#), then [check the answer key](#).